

also feeds on unripe green fruits and leaves large bites on the fruit leading to plant pathogens. Farmers view this moth species as destructive insects, so they use various methods to kill this insect.

The researchers of UW place these moths in a refrigerator or on ice for 15 minutes to anaesthetise and remove one antenna, which plays the role of a sensor in their smellicopter device. The removed antenna is biologically and chemically active for four hours with a continuous decrease in its signal strength.

Flying UAVs are used for military services, in space research, in searching and rescuing lost people, in sensing leakage of gas in an environment, in tracking wildlife populations, and much more. The UW research team has compared their smellicopter with the other miniature drones that use semiconductor sensors for chemical sensing, and have reported that their hybrid biological nanodevice can detect the trail of a volatile organic chemical within a minute whereas the other similar devices require around ten minutes.

Stainless steel wires inserted to both the base of the antenna and the EAG circuit can produce a voltage ranging from $10\mu\text{V}$ to 1mV in response to any chemical stimuli. The EAG sensor is interfaced with the smallest autonomous nanodrone Crazyflie 2.0, which occupies an area of only 85cm^2 . Crazyflie comes with its own external sensor that uses an optical flow camera and infrared laser range finder that helps the device in measuring the velocity, hovering without a GPS and without drifting. Though insects can do 3D tracking, the team has tried 2D cast-and-surge algorithm that requires upwind flying to test the proficiency of odour localisation, and many other strategies are yet to be tested in their future work.

In a fairly consistent wind, smellicopter takes off at a height of 40cm and hovers for ten seconds, casts left and right crosswind, avoiding any obstacles coming on its way. When there is a chemical plume, the device will heave 25cm upwind and then resume casting. The team has added two thin plastic vanes that con-

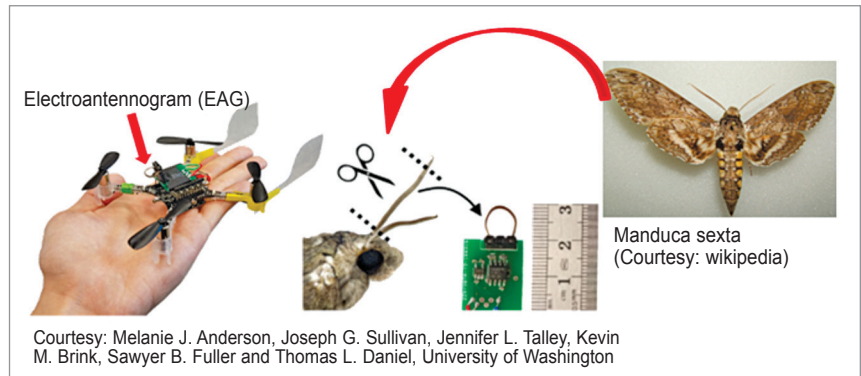


Fig. 2: Elements of a smellicopter

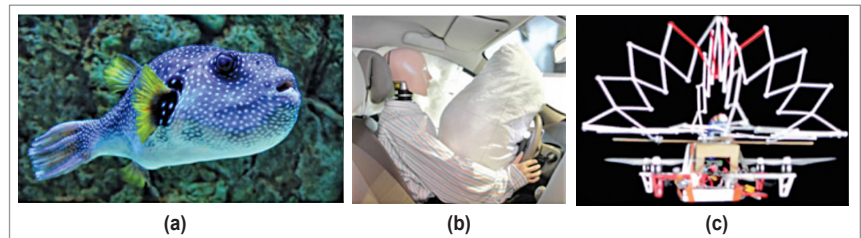


Fig. 3: (a) Pufferfish (Courtesy: starkrusher, flickr), (b) vehicle airbag (Courtesy: Vereshchagin Dmitry, Shutterstock), (c) Pufferbot

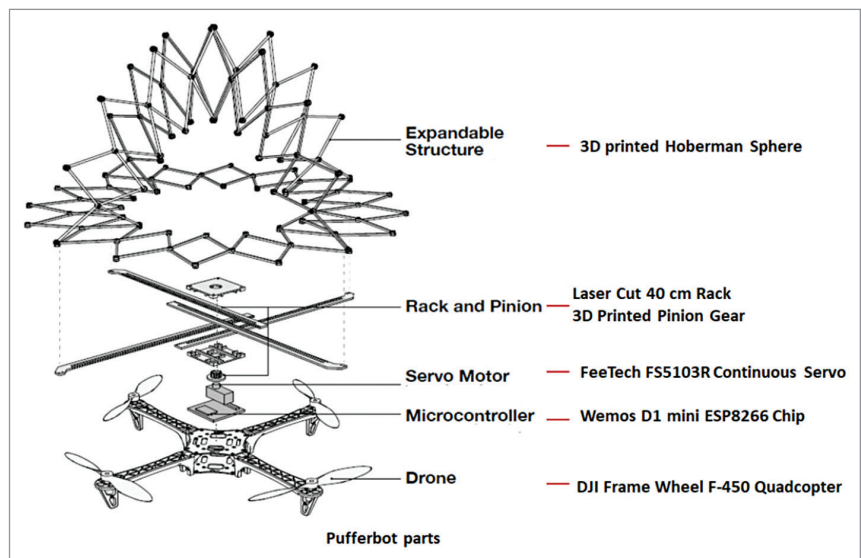


Fig. 4: Parts of a Pufferbot

stantly face upwind. In times when the device is not facing upwind, the force of the wind on the vanes will rotate the vehicle to face the upwind. The team is aiming to develop a smellicopter that can detect carbon-dioxide and chemical traces from an unexploded device.

Pufferbot—an aerial robot that expands to protect propellers

We know how pufferfish expands to an enormous size when threatened so

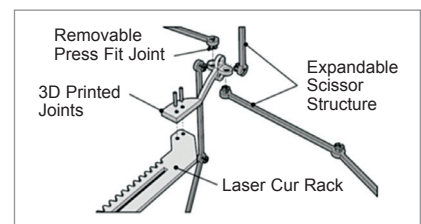


Fig. 5: Assembly of the Pufferbot

the predator won't be able to get their mouth on it, and the instant inflation of car airbags in the event of a crash.